Dissecting Transformers: A CLEAR Perspective Towards Green AI

Anonymous ACL submission

1 H100 Hardware Details

The NVIDIA H100 is based on the Hopper microarchitecture featuring fourth-generation Tensor Cores with a Transformer Engine enabling native FP8 inference. The H100 SXM5 variant provides 16,896 CUDA cores, 528 Tensor Cores, 80 GB of memory over a 5,120-bit interface ($\sim$3.35 TB/s bandwidth), and a total board power of 700 W. As with the Ada GPUs, the precise NVML power-sampling interval for the H100 is not published by NVIDIA. However, due to faster computation, we increase the number of repetitions to 20,000 to capture the fine-grained energy consumption precisely.

2 Experimental Insights for Hopper (H100) Architecture

To account for the generalizability of results and to analyse the impact of GPU architecture, we perform additional energy analysis on H100 Hopper architecture. We perform experiments on 4 decoder-only models of similar sizes(3-4B) namely, Llama-3.2-3B-Instruct, Qwen2.5-3B-Instruct, Phi-4-mini-instruct and Gemma-3-4B-it. We ran four family of energy measurements on H100: 1 Batch-size energy analysis at fixed sequence length of 32 covering batch sizes from 1 to 64 across MLP, Attention, Full Model and LM Head blocks. 2 Impact of KV Cache by measuring no-cache vs prefilled-cache energy of attention and full-model forwards across 4 models. 3 per-component Energy-GLOPs analysis and how the trends differ between A6000 and H100 architectures. 4 Attention variants and optimizations across 7 different Attention implementations as in the Main Draft. Here, the Main Draft refers to the original ARR submitted draft.

1 Batch Size

We capture the energy across 4 models for Attention, MLP, LM Head and Full model to see the impact of batching on inference. We use Energy per sample (mJ/sample) to compare the impact,to compare the impact, as it captures the reduction in energy consumption for each individual sample processed and output generated. We observe that increasing the batch size alone leads to about 42-48% drop in energy consumption. LM Head and MLP saturate energy gains (about 82% energy reduction) at batch size 16 and only an increase of 0.7 percentage points(pp) is achieved at batch size 64, however, Attention keeps shedding energy past a batch size of 16, reducing energy consumption by 4-5% at batch size 64 and reaching about 95-97% energy reduction. Gemma's per-block fixed-shape is heavier with 4 layernorms instead of 2, and the dual-rotary path computes both full and sliding attention rotary tables for every layer, due to which Gemma-3-4B's batch size as 1 full-model energy is 70% higher than Llama-3.2-3B's energy and stays elevated at every batch size. Refer Figure 1

2 KV Cache

We perform energy measurements for 4 decoder models, for the Attention block and full model to capture the energy savings by KV cache. Essentially we observe that for Attention blocks of Qwen2.5-3B and Gemma-3-4B, the energy savings due to KV Cache are comparatively lesser (about 29-34% at 512 tokens) than Llama3.2-3B and Phi-4-mini (about 46-51% at 512 tokens). Full model energy drops are around 52-54% for Llama3.2-3B and Phi-4-mini whereas it is much lesser (34% and 46%) for the other 2 models. Refer Table 1

3 Energy per FLOPs

On Energy per FLOP analysis we observe that component-wise comparison of mJ/GFLOP is preserved between the two GPUs: Attention is most expensive per FLOP, followed by MLP and LM Head. We empirically observe a significant change in the spread for both the GPUs. On A6000 the gap between MLP and Attention for $E/FLOPS$

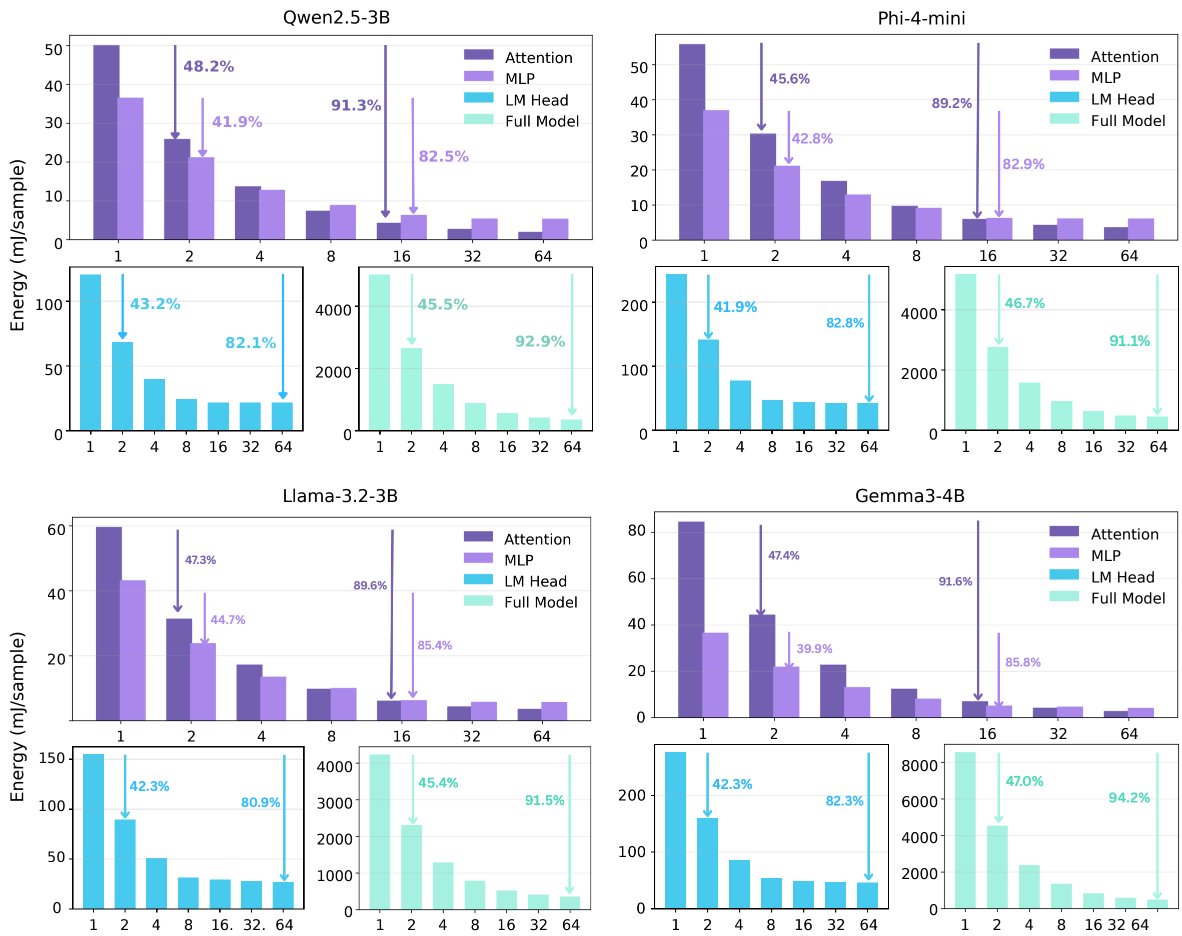

Figure 1: Energy per sample (Y-axis) comparison for different components across 4 models and the captured reduction in energy across batch sizes(X-axis). We use input sequence length 32 across energy measurements.

is around 2.7× times (8.4 vs 3.1 mJ/GFLOP in Llama-3.2-3B) , whereas on H100 it widens to 5.4× times (10.8 vs 2.0 mJ/GFLOP). As H100 makes the dense-GEMM components much cheaper per FLOP — its FP16 tensor-core uplift over A6000 is largest exactly where utilisation is already highest. Attention does not benefit symmetrically because its kernel is memory-bandwidth bound at moderate S. H100's faster compute can't help when the bottleneck is HBM. Refer Figure 2

4 Attention Variants and Optimizations

For AdaLovelace (A6000) and Hopper (H100) architectures, we observe a same trend for different variants and optimizations across all 4 decoder-only models. The Optimization tier i.e. Eager Attention with torch.compile(mode="max-autotune" or "reduce-overhead") consumes about 3 to 5 times lesser energy than Eager Attention and about 2.5 times lesser energy than Flash Attention across models. Refer Table 3 for precise energy measurements.

Component	8 Tok.	16 Tok.	32 Tok.	64 Tok.	128 Tok.	256 Tok.	512 Tok.
Qwen2.5-3B							
Attn. w/o KV Cache	49.98	50.21	50.77	52.88	55.30	61.41	77.02
Attn. w/ KV Cache	50.12	49.95	49.62	50.26	50.83	50.51	50.68
% Reduction (Attn.)	-0.29	0.53	2.27	4.96	8.08	17.76	34.19
Full Model w/o KV Cache	4543.1	4616.1	4838.6	5206.2	5692.8	6886.0	9247.9
Full Model w/ KV Cache	4623.3	4606.0	4700.8	4776.8	4680.4	4720.4	4902.7
% Reduction (Full Model)	-1.77	0.22	2.85	8.25	17.78	31.45	46.99
Llama-3.2-3B							
Attn. w/o KV Cache	56.43	57.37	58.16	60.93	66.71	79.36	101.91
Attn. w/ KV Cache	55.70	56.70	56.81	55.85	56.18	55.23	54.65
% Reduction (Attn.)	1.29	1.16	2.33	8.34	15.79	30.40	46.37
Full Model w/o KV Cache	3811.0	3906.3	4077.1	4404.2	4890.6	5994.3	8151.5
Full Model w/ KV Cache	3830.3	3807.5	3894.9	3885.8	3905.0	3914.3	3843.2
% Reduction (Full Model)	-0.50	2.53	4.47	11.77	20.15	34.70	52.85
Phi-4-mini							
Attn. w/o KV Cache	51.06	52.28	53.11	55.80	62.19	73.78	98.52
Attn. w/ KV Cache	48.65	49.67	50.39	49.08	49.54	48.84	47.91
% Reduction (Attn.)	4.72	4.99	5.12	12.04	20.34	33.80	51.37
Full Model w/o KV Cache	4196.7	4338.8	4454.7	4791.0	5386.6	6709.9	9501.9
Full Model w/ KV Cache	4126.3	4205.8	4210.5	4222.5	4196.1	4237.4	4353.9
% Reduction (Full Model)	1.68	3.06	5.48	11.87	22.10	36.85	54.18
Gemma-3-4B							
Attn. w/o KV Cache	81.33	83.89	87.56	90.06	94.61	103.86	119.45
Attn. w/ KV Cache	82.00	82.72	85.00	85.59	82.47	80.67	84.88
% Reduction (Attn.)	-0.83	1.40	2.92	4.96	12.83	22.33	28.94
Full Model w/o KV Cache	7728.2	7787.8	8193.1	8454.9	8939.6	10098.4	12176.9
Full Model w/ KV Cache	7855.5	7987.3	8102.3	7925.2	7897.6	7935.4	8026.5
% Reduction (Full Model)	-1.65	-2.56	1.11	6.26	11.66	21.42	34.08

Table 1: KV Cache energy analysis on NVIDIA H100 GPU for decoder-only models (Qwen2.5-3B, Llama-3.2-3B, Phi-4-mini, Gemma-3-4B)for the Attention (Attn.) block and Full Model with (w/) and without (w/o) KV Cache across input sequence lengths of 8–512 tokens. Energy values are mean measurements in millijoules (mJ) across 20 trial. % Reduction denotes the percentage reduction in energy when using KV cache. Negative values indicate a slight overhead at short sequences where KV cache provides no benefit.

Component	BS=1	BS=2	BS=4	BS=8	BS=16	BS=32	BS=64
Qwen2.5-3B							
MLP	36.60	21.25	12.85	8.96	6.39	5.47	5.43
Attention	50.16	25.96	13.76	7.48	4.36	2.79	2.02
LM Head	120.26	68.31	39.58	24.34	21.67	21.83	21.54
Full Model	5008.58	2650.60	1488.52	879.21	570.07	425.44	354.13
Llama-3.2-3B							
MLP	43.26	23.91	13.58	10.09	6.33	5.84	5.77
Attention	59.68	31.45	17.30	9.85	6.18	4.42	3.65
LM Head	154.72	89.25	50.63	31.05	29.44	27.56	26.91
Full Model	4234.68	2312.24	1291.55	791.72	523.70	412.54	359.47
Phi-4-mini							
MLP	37.05	21.21	13.01	9.20	6.33	6.19	6.17
Attention	55.84	30.36	16.88	9.75	6.04	4.35	3.69
LM Head	243.52	141.43	77.55	47.16	43.70	42.42	42.01
Full Model	5178.49	2761.60	1579.09	962.98	626.60	492.71	459.59
Gemma-3-4B							
MLP	36.69	22.04	13.20	8.22	5.21	4.76	4.22
Attention	84.67	44.51	22.89	12.51	7.08	4.29	2.86
LM Head	276.10	159.28	85.39	53.77	48.37	46.77	46.09
Full Model	8562.91	4538.00	2387.06	1366.49	839.99	595.12	496.81

Table 2: Per-sample energy consumption (mJ/sample) on NVIDIA H100 GPU across batch sizes(BS) 1–64 for decoder-only models (Qwen2.5-3B, Llama-3.2-3B, Phi-4-mini, Gemma-3-4B) for different components. Increasing batch size amortises fixed GPU costs, yielding substantial per-sample energy reductions. Values represent mean per-sample energy in millijoules (mJ).

Attention Variant	Energy (mJ)	Std. Dev.	FLOPs (GFLOPs)	Energy (mJ)/GFLOP
Qwen2.5-3B				
FP16 Flash Attention	74.707	1.586	0.604	123.645
FP16 Eager Attention	86.236	1.298	0.613	140.769
FP16 SDPA	59.473	1.073	0.604	98.433
FP16 Flash w/ torch.compile()	48.385	1.173	0.612	79.012
FP16 Eager w/ torch.compile()	47.883	1.179	0.612	78.193
BF16 Flash Attention	74.541	1.099	0.604	123.372
FP16 Eager Max Autotune	17.149	1.047	–	–
FP16 Eager Reduce Overhead	16.805	1.053	–	–
Llama-3.2-3B				
FP16 Flash Attention	82.488	0.683	1.611	51.203
FP16 Eager Attention	92.737	1.421	1.624	57.118
FP16 SDPA	67.465	1.208	1.611	41.878
FP16 Flash w/ torch.compile()	54.342	1.162	1.623	33.479
FP16 Eager w/ torch.compile()	54.942	1.163	1.623	33.848
BF16 Flash Attention	81.563	1.286	1.611	50.628
FP16 Eager Max Autotune	23.406	0.837	–	–
FP16 Eager Reduce Overhead	22.496	0.932	–	–
Phi-4-mini				
FP16 Flash Attention	78.502	1.536	1.611	48.732
FP16 Eager Attention	85.605	0.736	1.624	52.728
FP16 SDPA	64.069	1.496	1.611	39.772
FP16 Flash w/ torch.compile()	45.176	1.240	1.623	27.831
FP16 Eager w/ torch.compile()	43.858	1.216	1.623	27.020
BF16 Flash Attention	77.698	0.760	1.611	48.232
FP16 Eager Max Autotune	24.137	1.487	–	–
FP16 Eager Reduce Overhead	22.822	1.485	–	–
Gemma-3-4B				
FP16 Flash Attention	120.735	0.767	1.007	119.881
FP16 Eager Attention	129.189	1.899	1.016	127.215
FP16 SDPA	100.528	1.073	1.007	99.817
FP16 Flash w/ torch.compile()	51.339	0.741	1.015	50.579
FP16 Eager w/ torch.compile()	50.090	0.986	1.015	49.349
BF16 Flash Attention	121.200	0.712	1.007	120.342
FP16 Eager Max Autotune	18.915	1.285	–	–
FP16 Eager Reduce Overhead	19.164	1.070	–	–

Table 3: Average energy consumption, FLOPs, and energy efficiency (mJ/GFLOP) of attention variants and optimisations on NVIDIA H100 GPU for Qwen2.5-3B, Llama-3.2-3B, Phi-4-mini, and Gemma-3-4B at 32 input tokens (layer index 14). `torch.compile` modes Max Autotune and Reduce Overhead do not report FLOPs (–). Mean energy and standard deviation are in millijoules (mJ).

Component	1 Tok.		8 Tok.		32 Tok.		64 Tok.		128 Tok.		256 Tok.		512 Tok.	
	Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.
Qwen2.5-3B														
Attention Block	51.541	0.596	52.284	0.487	56.799	1.042	53.857	0.678	57.726	0.882	67.171	0.665	82.680	1.023
MLP	36.154	0.658	37.808	0.990	43.312	1.001	46.869	1.115	59.372	0.310	81.394	1.714	103.319	1.335
Norm. (All)	22.630	–	23.074	–	23.654	–	23.365	–	23.806	–	26.516	–	29.250	–
Captured (Block)	110.325	–	113.166	–	123.765	–	124.091	–	140.904	–	175.081	–	215.249	–
Block	126.115	2.006	128.354	1.760	140.730	1.626	141.221	1.296	156.666	1.909	192.985	2.302	254.608	2.936
%Capture (Block)	87.48	–	88.17	–	87.95	–	87.87	–	89.94	–	90.72	–	84.54	–
Final Layer Norm	11.667	0.504	11.377	0.519	11.721	0.376	11.339	0.627	12.038	0.689	13.270	0.629	14.422	0.102
Embedding Layer	2.024	0.619	2.093	0.865	2.064	0.629	2.247	0.598	2.095	0.691	1.991	0.695	2.415	0.850
LM Head	107.811	1.841	113.887	1.752	128.718	2.075	148.488	1.857	162.839	2.558	199.686	2.049	358.296	1.728
Captured (Model)	4661.642	–	4748.101	–	5208.783	–	5246.030	–	5816.948	–	7162.407	–	9541.021	–
Model	5023.149	35.121	5144.147	31.225	5318.215	24.897	5600.742	42.397	6326.814	90.194	7575.968	68.096	9923.387	76.019
%Capture (Model)	92.80	–	92.30	–	97.94	–	93.67	–	91.94	–	94.54	–	96.14	–
Llama-3.2-3B														
Attention Block	56.940	0.652	57.227	0.705	59.390	0.714	62.943	0.670	69.822	0.817	83.120	1.351	108.452	1.597
MLP	36.309	0.884	37.399	0.955	43.127	1.037	50.069	1.147	57.397	1.358	81.234	1.325	99.442	2.315
Norm. (All)	21.695	–	22.628	–	23.388	–	23.403	–	24.660	–	26.204	–	31.213	–
Captured (Block)	114.944	–	117.254	–	125.905	–	136.415	–	151.879	–	190.558	–	239.107	–
Block	129.109	1.634	131.108	2.145	138.576	1.731	148.175	1.247	167.213	1.780	206.436	6.670	273.574	5.038
%Capture (Block)	89.03	–	89.43	–	90.85	–	92.06	–	90.83	–	92.31	–	87.40	–
Final Layer Norm	11.067	0.476	11.473	0.594	11.665	0.584	11.783	0.498	11.919	0.386	12.938	0.045	14.992	0.691
Embedding Layer	1.676	0.585	1.823	0.606	2.128	0.570	2.077	0.624	2.212	0.659	2.065	0.722	2.893	0.911
LM Head	134.341	1.727	139.279	1.875	159.250	2.042	184.091	2.544	201.910	0.327	246.629	2.840	470.092	0.876
Captured (Model)	3762.136	–	3823.599	–	4053.171	–	4346.851	–	4898.005	–	6041.840	–	8148.049	–
Model	3982.171	14.261	4069.615	11.394	4309.644	11.743	4621.326	7.462	5193.840	41.648	6383.030	84.892	8592.957	90.873
%Capture (Model)	94.47	–	93.95	–	94.05	–	94.06	–	94.31	–	94.65	–	94.82	–
Phi-4-mini														
Attention Block	49.427	0.932	52.847	0.800	54.815	0.830	57.805	0.835	64.555	0.841	78.723	1.230	106.575	1.730
MLP	32.453	0.852	34.327	1.225	39.073	0.883	42.827	1.134	54.367	1.771	76.341	1.946	104.180	2.126
Norm. (All)	20.632	–	20.575	–	21.445	–	21.681	–	23.773	–	26.247	–	30.708	–
Captured (Block)	102.512	–	107.749	–	115.333	–	122.313	–	142.695	–	181.311	–	241.463	–
Block	121.446	2.152	122.747	1.966	131.879	2.808	141.290	2.169	157.501	4.549	200.226	2.815	275.316	3.290
%Capture (Block)	84.41	–	87.78	–	87.46	–	86.57	–	90.60	–	90.55	–	87.71	–
Final Layer Norm	10.274	0.538	10.387	0.540	10.645	0.549	10.872	0.600	12.090	0.755	13.074	0.516	15.387	0.655
Embedding Layer	1.995	0.537	1.779	0.591	1.840	0.610	2.049	0.619	2.184	0.662	2.260	0.728	3.116	0.820
LM Head	207.887	0.542	214.748	1.683	248.699	2.630	284.385	2.722	308.104	2.095	373.508	2.330	694.144	2.122
Captured (Model)	4106.428	–	4154.818	–	4481.312	–	4818.586	–	5362.410	–	6796.074	–	9522.759	–
Model	4569.910	14.540	4707.764	21.491	4963.441	22.165	5318.060	13.590	6059.853	47.785	7444.260	6.646	10287.824	29.543
%Capture (Model)	89.86	–	88.26	–	90.29	–	90.61	–	88.49	–	91.29	–	92.57	–
Gemma-3-4B														
Attention Block	80.871	0.869	83.009	1.163	86.033	1.163	86.785	0.676	95.199	1.734	103.681	2.098	119.920	2.765
MLP	37.461	0.889	38.904	1.015	41.080	1.127	46.252	1.256	54.374	0.994	70.576	1.406	92.175	2.099
Norm. (All)†	51.811	–	52.959	–	54.663	–	54.889	–	56.953	–	60.866	–	67.851	–
Captured (Block)	170.143	–	174.872	–	181.776	–	187.926	–	206.526	–	235.123	–	279.946	–
Block	185.514	3.147	195.937	1.830	201.324	2.437	206.608	1.588	219.373	1.328	254.237	1.818	302.048	2.405
%Capture (Block)	91.71	–	89.25	–	90.29	–	90.96	–	94.14	–	92.48	–	92.68	–
Final Layer Norm	13.401	0.358	13.343	0.019	13.572	0.017	13.408	0.553	14.122	0.029	15.222	0.542	16.887	0.413
Embedding Layer	3.364	0.479	4.104	0.588	3.643	0.662	3.840	0.009	4.289	0.396	4.665	0.603	6.353	1.004
LM Head	234.403	2.346	235.794	1.611	248.612	2.276	290.266	3.245	300.598	3.061	388.972	2.070	598.837	4.466
Captured (Model)	6558.644	–	6915.099	–	7110.843	–	7332.186	–	7777.691	–	9052.917	–	10891.709	–
Model	8018.656	33.503	8094.162	8.659	8277.355	66.399	8535.255	52.762	9127.005	48.135	10473.400	35.654	12651.708	67.154
%Capture (Model)	81.80	–	85.43	–	85.91	–	85.91	–	85.21	–	86.44	–	86.09	–

Table 4: Energy of decoder-only model components using CLEAR on NVIDIA H100 GPU (Qwen2.5-3B, Llama-3.2-3B, Phi-4-mini, Gemma-3-4B) across input sequence lengths 1-512

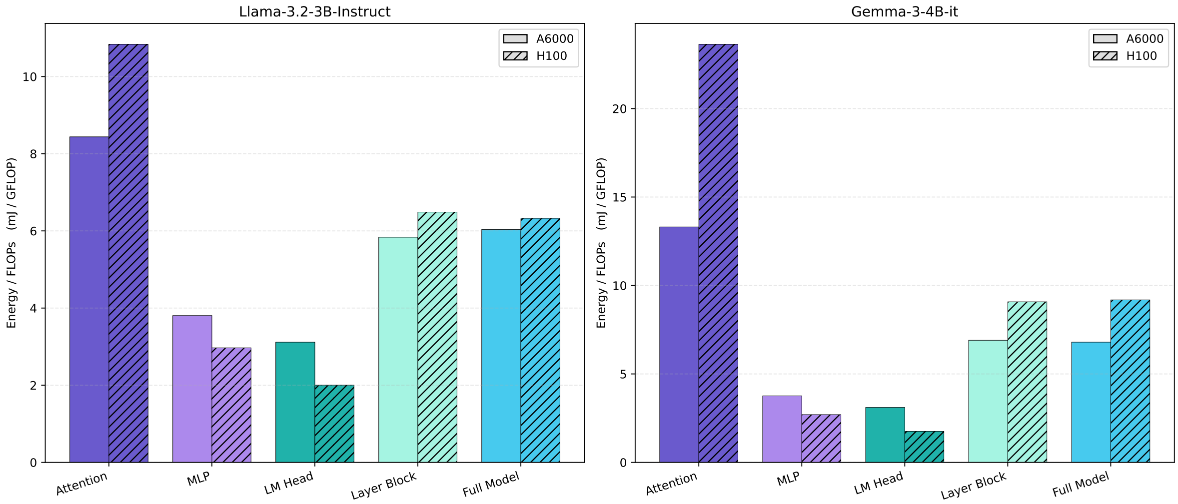

Figure 2: Energy per FLOPs analysis for the Llama-3.2-3B and Gemma-3-4B-it models across Attention, MLP, LM-Head, Layer Block and Full Model. All measurements are done using a sequence length of 128.

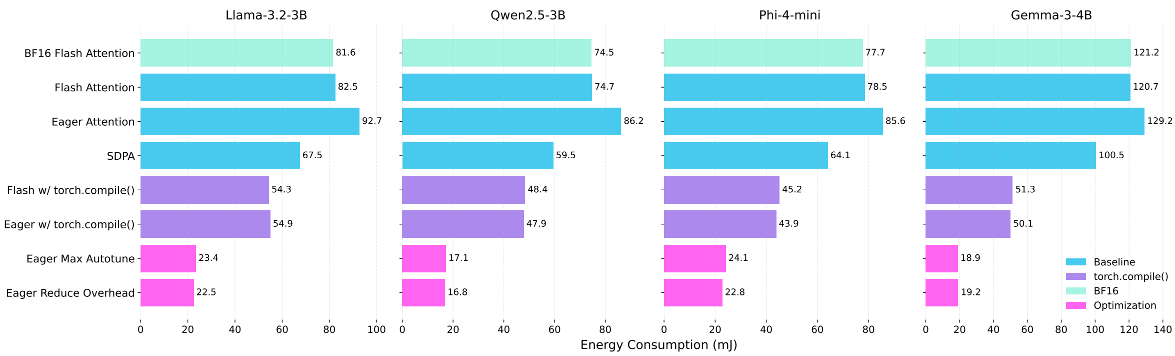

Figure 3: Energy consumption of energy block for different Attention Variants and Optimizations for sequence length 128.